

Profluent Successfully Edits Human Genome with OpenCRISPR-1, the World's First AI-Created and Open-Source Gene Editor

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BERKELEY, Calif.--(BUSINESS WIRE)--Profluent, the AI-first protein design company, today debuted the OpenCRISPR™ initiative, releasing the world's first open-source, AI-generated gene editor. With this launch, Profluent demonstrates the first successful precision editing of the human genome with customizable gene editors designed from scratch with AI.

OpenCRISPR-1 is an AI-created gene editor, consisting of a Cas9-like protein and guide RNA, fully developed using Profluent's large language models (LLMs). Through the training process for OpenCRISPR, the company's AI learned from massive scale sequence and biological context to generate millions of diverse CRISPR-like proteins that do not occur in nature, thereby exponentially expanding virtually all known CRISPR families.

In its commitment to democratizing the technology, Profluent has launched OpenCRISPR-1 as an initial open-source release, making the AI-designed gene editor freely available to license for ethical research and commercial uses.

"Attempting to edit human DNA with an AI-designed biological system was a scientific moonshot," said Ali Madani, Profluent co-founder and Chief Executive Officer. "Our success points to a future where AI precisely designs what is needed to create a range of bespoke cures for disease. To spur innovation and democratization in gene editing, with the goal of pulling this future forward, we are open-sourcing the products of this initiative."

Profluent's hope is that CRISPR medicines become available to more patients who may benefit, and for a broader range of disorders. The unique power of AI is that it enables researchers to reimagine and build gene editing systems from the ground up, which would be impossible with conventional protein engineering methods. By growing the universe of CRISPR families and releasing foundational molecules that can be further built upon, this application of AI paves the way to greater access and lower costs for gene editing therapies.

"It's phenomenal that the first CRISPR-based treatments for genetic diseases such as sickle cell disease are already changing the lives of patients, but there remains an urgent need to accelerate the development of this technology for thousands of other currently incurable diseases," said Hilary Eaton, Chief Business Officer at Profluent. "Our intention with

OpenCRISPR is to partner with cutting-edge research institutions and drug developers with a powerful and practical way to safely expedite the development of new CRISPR genetic therapies.”

“Today’s announcement is a watershed moment and the beginning of what we hope will be an iterative process as we embark on this next generation of building genetic medicines,” said Peter Cameron, Vice President and Head of Gene Editing at Profluent. “We encourage the gene editing community to pressure test OpenCRISPR-1. If there are particular features that could be improved for a specific application, we’d like to know and can collaborate to optimize those properties.”

Profluent has published the science behind its OpenCRISPR initiative in a preprint publication. In adherence to ethical standards, the company's OpenCRISPR license will contain certain excluded uses, such as human germline editing. For more details on OpenCRISPR, visit <https://www.profluent.bio>.

About Profluent

Profluent is an AI-first protein design company headquartered in Berkeley, CA. Founded in 2022, Profluent develops deep generative models to design and validate novel, functional proteins to revolutionize biomedicine. The company is backed by leading investors including Spark Capital, Insight Partners, Air Street Capital, AIX Ventures, and Convergent Ventures. To learn more about Profluent’s mission to decode the language of life with AI, visit [profluent.bio](https://www.profluent.bio).